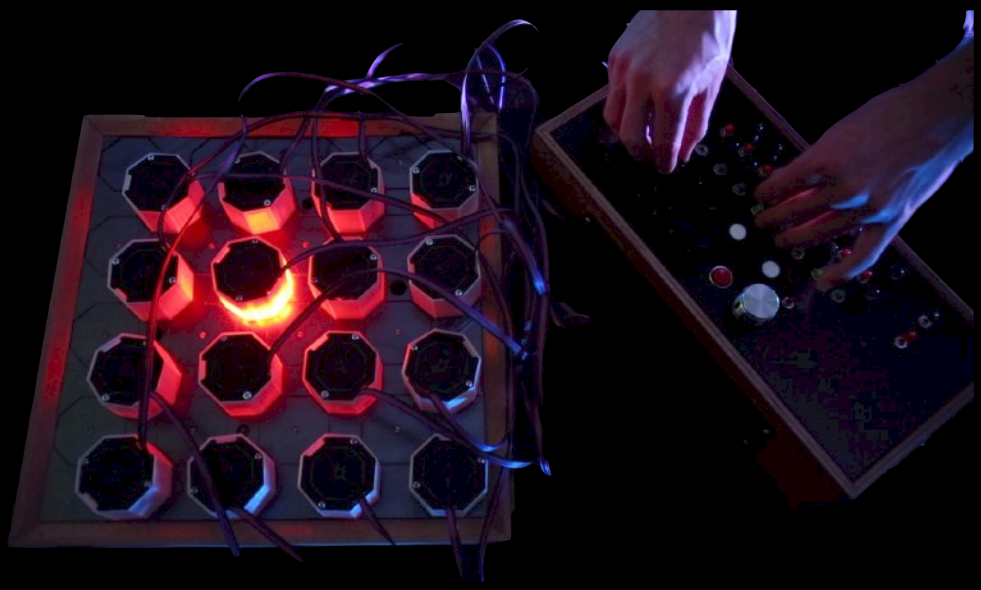
# LSe - Sculpting Sound with Light



LSe (Light-Sequencer) is an award-winning system designed specifically for live electronic music performance and interaction. It allows users to create sequences of light, that are then used to control, structure and interact with various parameters of sound. One of the sources of inspiration for this project was the issue of, 'liveness' of electronic music performance. How does the audience know if the performance is really live? With the 'LSe' every sound is sculpted in real time using light, creating a direct link between the visual sonic realm.

## Technical Overview

**Built with Raspberry Pi, Pure Data and Arduino**

**Contains 3 physical modules**

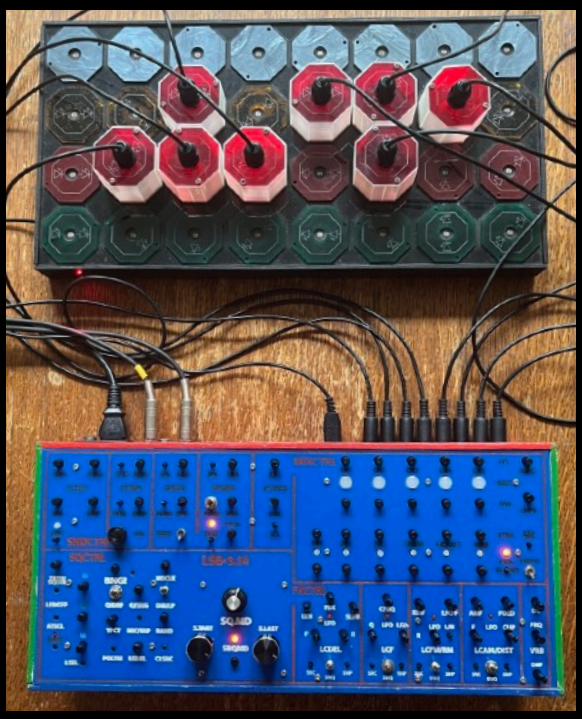
- **Sensor Panel:** 32 LDRs
- **Light Sequencer:** 8 smart RGB LEDs
- **Control Panel** with 4 sections: Mix, FX, Snd, Seq

**Sound engine**

- Full-duplex audio I/O via Audio Injector sound card
- Dynamic patch swapping: sound and FX patches can be swapped
- Three light-controlled sound patches
- 32 voices Polyphony
- Three send FX patches per generator and external audio input

**Light Sequencer**

- 6 sequencer modes with 15 sub-modes
- LEDs can work as one sequence or as eight independent sequences
- Clock modes: Manual, Auto, Regular Random, Swing
- Clock sources: Internal, Tap Tempo, Wireless Sync via Ableton Link



- Currently available sound patches:
  - Subtractive: 32-band 3rd-octave filter bank driven by noise and feedback
  - Additive: 32 oscillators in configurable harmonic ratios
  - Sample-based patch I – granular time-stretch sampler
  - Sample-based patch II – looper / sampler mangler
  - Pinging resonator: resonant filter

- Currently available effects patches:
  - LCD: Light-Controlled Delay with dedicated LFO
  - LCF: Light-Controlled Variable State Filter with dedicated LFO
  - LCAM: Light-Controlled Amplitude Modulation/Distortion with dedicated LFO
  - LCFM: Light-Controlled Frequency Modulation with dedicated LFO
  - LCGV: Light-Controlled Granular Reverb

- Currently available Sequencer Modes and sub sequencer modes:
  - Classic mode: regular, random,  random wo repetition, random walk, chance 50/50
  - Poly mode: Modulo polyrhythm, euclidean,
  - Fade in and out
  - Tempo change with variable ramp
  - Poly 2: Phasing aka instant Steve Reich, 8 random paralel clocks, 8 divisions of a note,
  - Manual mode: manual control of each led independently

## Video examples:

Click on the Image to Watch the Video Teaser

Entire Performance (circa 18 min) at the Royal Conservatory, The Hague, NL

Entire Performance (circa 35 min) at Audio Art Festival Online Edition

Short Clip (circa 2 min)

If that isn't enough, here is a whole YouTube playlist with more examples

## Press Coverage

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